



Cardiovascular Pharmacology

Pharmacological evidence that spinal α_{2C} - and, to a lesser extent, α_{2A} -adrenoceptors inhibit capsaicin-induced vasodilatation in the canine external carotid circulation

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ARTICLE INFO

Article history:

Received 28 July 2011

Received in revised form 2 March 2012

Accepted 6 March 2012

Available online 14 March 2012

Keywords:

B-HT 933

BRL44408

CGRP

External carotid artery

JP-1302

Migraine

ABSTRACT

During a migraine attack capsaicin-sensitive trigeminal sensory nerves release calcitonin gene-related peptide (CGRP), producing cranial vasodilatation and central nociception; hence, trigeminal inhibition may prevent this vasodilatation and abort migraine headache. This study investigated the role of spinal α_2 -adrenoceptors and their subtypes (i.e. α_{2A} , α_{2B} and/or α_{2C} -adrenoceptors) in the inhibition of the canine external carotid vasodilator responses to capsaicin. Anaesthetized vagosympathectomized dogs were prepared to measure arterial blood pressure, heart rate and external carotid conductance. The thyroid artery was cannulated for one-min intracarotid infusions of capsaicin, α -CGRP and acetylcholine. A cannula was inserted intrathecally for spinal (C_1 – C_3) administration of 2-amino-6-ethyl-4,5,7,8-tetrahydro-6H-oxazolo-[5,4-d]-azepin-3-dihydrochloride (B-HT 933; a selective α_2 -adrenoceptor agonist) and/or the α_2 -adrenoceptor antagonists rauwolscine ($\alpha_{2A/2B/2C}$), 2-[(4,5-dihydro-1H-imidazol-2-yl)methyl]-2,3-dihydro-1-methyl-1H-isoindole maleate (BRL44408; α_{2A}), imiloxan (α_{2B}) or acridin-9-yl-[4-(4-methylpiperazin-1-yl)-phenyl]amine (JP-1302; α_{2C}). Infusions of capsaicin, α -CGRP and acetylcholine dose-dependently increased the external carotid conductance. Intrathecal B-HT 933 (1000 and 3100 μ g) inhibited the vasodilator responses to capsaicin, but not those to α -CGRP or acetylcholine. This inhibition, abolished by rauwolscine (310 μ g), was: (i) unaffected by 3100 μ g imiloxan; (ii) partially blocked by 310 μ g of BRL44408 or 100 μ g of JP-1302; and (iii) abolished by 1000 μ g of BRL44408 or 310 μ g of JP-1302. Thus, intrathecal B-HT 933 inhibited the external carotid vasodilator responses to capsaicin. This response, mediated by spinal α_2 -adrenoceptors unrelated to the α_{2B} -adrenoceptor subtype, resembles the pharmacological profile of α_{2C} -adrenoceptors and, to a lesser extent, α_{2A} -adrenoceptors.

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1. Introduction

Migraine is a complex and highly prevalent neurovascular disorder that may involve: (i) activation of the trigeminovascular complex (Holland, 2009); (ii) high circulating plasma concentrations of calcitonin gene-related peptide (CGRP) (Goadsby et al., 2002); (iii) vasodilatation of intracranial arteries and of the extracranial branches of the external carotid artery, which are a source of pain in migraine (Olesen et al., 2009; Shevel, 2011a, 2011b); and (iv) a wide dysfunction of the sensory system (Sprenger and Goadsby, 2009).

CGRP, which is mainly located on sensory neurons and perivascular nerves (Van Rossum et al., 1997), dilates cranial blood vessels and transmits vascular nociception (Arulmani et al., 2004). The release of endogenous CGRP can be experimentally induced by either trigeminal electrical stimulation (Goadsby and Edvinsson, 1993) or chemical

stimulation with capsaicin (Dux et al., 2003; Potenza et al., 1994), which activates perivascular sensory nerves through the stimulation of the transient receptor potential vanilloid type 1 (TRPV1) cation channel (Caterina et al., 1997; Kim et al., 2012). In this respect, we have shown that the external carotid vasodilatation produced by intracarotid capsaicin in anaesthetized dogs is: (i) unaltered after i.v. sumatriptan (Muñoz-Islas et al., 2006), which does not easily cross an intact blood-brain barrier (Humphrey et al., 1991); (ii) inhibited after spinal (C_1 – C_3) administration of sumatriptan via central 5-HT_{1B} receptors (Muñoz-Islas et al., 2009); and (iii) inhibited after i.v. administration of the traditional lipid-soluble α_2 -adrenoceptor agonist clonidine via MK912-sensitive $\alpha_{2A/2C}$ -adrenoceptors (Jiménez-Mena et al., 2006; Table 1). These results suggest that activation of central mechanisms can inhibit capsaicin-induced cranial vasodilatation, but it is not clear the extent to which peripheral and central $\alpha_{2A/2C}$ -adrenoceptors contribute to the above inhibition by i.v. clonidine. Indeed, besides producing peripheral inhibition on sensory perivascular CGRPergic neurons by $\alpha_{2A/2C}$ -adrenoceptors (Villalón et al., 2008), clonidine readily crosses the blood-brain barrier (McLennan, 1981; Van Zwieten et al., 1984) and, thus, it may exert central effects. Hence, an experimental

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Table 1

Binding affinity constants (pK_i) of several antagonists for cloned human α_2 -adrenoceptor subtypes.

Antagonist	α_2 -adrenoceptor subtype		
	α_{2a}	α_{2b}	α_{2c}
Rauwolscine	9.5 ^a	9.4 ^a	9.9 ^a
	8.9 ^b	8.9 ^b	9.3 ^b
BRL44408	8.2 ^c	6.2 ^c	6.8 ^c
	7.6 ^d	6.0 ^d	6.4 ^d
Imiloxan	5.8 ^d	6.9 ^d	6.0 ^d
	6.5 ^b	7.2 ^b	6.8 ^b
MK912	8.9 ^c	8.9 ^c	10.2 ^c
	9.1 ^b	9.1 ^b	10.2 ^b
JP-1302	5.5 ^e	5.8 ^e	7.5 ^e

^aData taken from: Bylund (1992). ^bData taken from: Jasper et al. (1998). ^cData taken from: Uhlén et al. (1994). ^dData taken from: Devedjian et al. (1994). ^eData taken from: Sallinen et al. (2007).

approach that would allow us to investigate the exclusive role of central (spinal) α_2 -adrenoceptors in the above inhibition of capsaicin-induced external carotid vasodilatation in dogs is the pharmacological analysis of the effects produced by spinal (C_1 – C_3) administration of a selective α_2 -adrenoceptor agonist, such as B-HT 933.

Since α_2 -adrenoceptors exist in three functional subtypes, i.e. α_{2A} , α_{2B} , and α_{2C} -adrenoceptors, (Bylund et al., 1994; Docherty, 1998), this study investigated in vagosympathectomized dogs: (i) whether direct spinal (C_1 – C_3) administration of B-HT 933 results in inhibition of the external carotid vasodilator responses to intracarotid capsaicin, α -CGRP and acetylcholine; and (ii) the role of spinal $\alpha_{2A/2B/2C}$ -adrenoceptor subtypes in the above inhibition to B-HT 933 by analyzing the effects to spinal (C_1 – C_3) administration of the antagonists rauwolscine ($\alpha_{2A/2B/2C}$), BRL44408 (α_{2A}), imiloxan (α_{2B}) and JP-1302 (α_{2C}) (Table 1). This is a neurovascular model of migraine that has been useful for the screening of compounds with potential antimigraine activity (Gupta and Villalón, 2010; Saxena and De Vlaam-Schluter, 1974; Shevel, 2011a; Villalón et al., 2002).

2. Materials and methods

2.1. General methods

Experiments were carried out in a total of 66 male mongrel dogs (body weight: 18–25 kg) that were anaesthetised with an i.v. bolus injection of sodium pentobarbitone (30 mg/kg), and additional amounts (1 mg/kg, i.v.) were provided when required throughout the experiment for maintaining the anaesthesia (Villalón et al., 1999). All dogs were intubated with an endotracheal tube and artificially ventilated with room air, using a model 5025 Ugo Basile pump (rate: 20 strokes/min; stroke volume: 13–16 ml/kg) as established by Kleinman and Radford (1964). A catheter was placed in the right femoral artery, connected to a Grass pressure transducer (P23XL, Quincy, MA, U.S.A.) for the recording of blood pressure. Another catheter was placed intrathecally (Yaksh et al., 2004) through the subarachnoid space in the spinal C_1 – C_3 region for the administration of vehicle (saline), B-HT 933, rauwolscine, BRL44408, imiloxan or JP-1302. Since the trigeminal sensory nerve projects to the trigeminal nucleus caudalis in the medulla oblongata, the intrathecal administration of these compounds at the cervical level was carried out in an attempt to facilitate their diffusion to the medulla oblongata (with their resulting effects on perivascular trigeminal sensory terminals). After administration of compounds, the intrathecal catheter was flushed with 0.3 ml of saline. Mean arterial blood pressure (MAP) was calculated from the systolic (SAP) and diastolic (DAP) arterial pressures as follows: $MAP = DAP + (SAP - DAP) / 3$. Heart rate was measured with a tachograph (7P4F, Grass Instrument Co., Quincy, MA, U.S.A.) triggered from the blood pressure signal.

The right common carotid artery was dissected free and the corresponding internal carotid and occipital branches were ligated; under these experimental conditions, the blood flow through the right common carotid artery was considered to represent that of the external carotid artery; for further considerations see Villalón et al. (1993). Subsequently, an ultrasonic flow probe (4 mm R-Series) connected to an ultrasonic T206 flow meter (Transonic Systems Inc., Ithaca, NY, U.S.A.) was placed around the right common carotid artery. Bilateral cervical vagosympathectomy was systematically performed in order to prevent possible baroreceptor reflexes produced by the intracarotid continuous infusions of compounds. Thereafter, a catheter was introduced into the right cranial thyroid artery for the intracarotid continuous infusions of capsaicin, α -CGRP and acetylcholine.

It is to be noted that after bilateral vagosympathectomy the carotid arterioles are dilated under our experimental conditions (Villalón et al., 1995). Since the main objective of our study is to investigate the responses to vasodilator agents (such as capsaicin, α -CGRP and acetylcholine) on the external carotid circulation, we had to produce a selective carotid precontraction with an intracarotid continuous infusion of phenylephrine (an α_1 -adrenoceptor agonist); this procedure allowed us to enhance the vasodilator responses to capsaicin, α -CGRP and acetylcholine, as previously reported (Jiménez-Mena et al., 2006; Muñoz-Islas et al., 2006, 2009). For this purpose, a 0.5 mm (external diameter) needle, connected to a suitable catheter, was inserted into the right common carotid artery for the continuous infusion of phenylephrine by another motor driven syringe. Then, the vasodilator responses to capsaicin, α -CGRP and acetylcholine during the infusion of phenylephrine were compared with those observed after intrathecal administration of B-HT 933, antagonists, or equivalent volumes of vehicle. Capsaicin, α -CGRP, acetylcholine and phenylephrine were infused into the carotid artery by using WPI model sp100i pumps (World Precision Instruments Inc., Sarasota, FL, U.S.A.); for further details, see the experimental protocols described below.

Arterial blood pressure, heart rate and external carotid blood flow were recorded simultaneously by a model 7D polygraph (Grass Instrument Co., Quincy, MA, U.S.A.). The body temperature of the animals was maintained between 37 and 38 °C.

2.2. Experimental protocols

After the animals ($n=66$) had been in a stable haemodynamic condition for at least 60 min, baseline values of mean arterial blood pressure, heart rate and external carotid conductance were determined. Subsequently, all animals received an intracarotid continuous infusion of phenylephrine (1.5–6.8 μ g/min) to get a sustained carotid vasoconstriction (a 40–50% decrease in external carotid blood flow) (Jiménez-Mena et al., 2006; Muñoz-Islas et al., 2006). Twenty minutes later all the different treatments were given without interrupting the infusion of phenylephrine. Under these experimental conditions, the animals were divided into 4 groups ($n=14$, 8, 16, and 28).

The first group ($n=14$) received consecutive intracarotid infusions (1 ml/min, for 1 min) of capsaicin (10, 18, 31 and 56 μ g/min) (control curve). Then, this group was subdivided into 3 subgroups that received intrathecal bolus injections (0.5 ml) of, respectively: (i) vehicle (saline; given twice, $n=4$); (ii) B-HT 933 (310 and 1000 μ g given cumulatively; $n=6$); and (iii) B-HT 933 (1000 and 3100 μ g given cumulatively; $n=4$). Ten minutes after intrathecal administration of each dose of vehicle or B-HT 933, a dose–response curve to capsaicin was elicited again; this was completed 30 min after starting the infusion of the first dose of capsaicin. The next dose of B-HT 933 or its equivalent volume of vehicle was given when the administration scheme of capsaicin had been finished.

The second group ($n=8$) received consecutive intracarotid infusions (1 ml/min, for 1 min) of capsaicin (10, 18, 31 and 56 μ g/min),

α -CGRP (0.1, 0.31, 1 and 3.1 $\mu\text{g}/\text{min}$) and acetylcholine (0.01, 0.03, 0.1 and 0.31 $\mu\text{g}/\text{min}$) (control curves). Subsequently, this group was subdivided into 2 subgroups ($n = 4$ each) that received intrathecal bolus injections (0.5 ml) of: (i) vehicle (saline); or (ii) 3100 μg B-HT 933 (which induced the maximum inhibition on the responses to capsaicin in the first group). Ten minutes later, the responses to the above infusions of capsaicin, α -CGRP and acetylcholine were elicited again.

The third group ($n = 16$) was subdivided into 5 subgroups that received intracarotid infusions of the above doses of capsaicin before (control curve) and 10 min after intrathecal bolus injections (0.5 ml) of, respectively: (i) vehicle (saline; $n = 4$); (ii) rauwolsine (310 μg ; $n = 3$); (iii) BRL44408 (1000 μg ; $n = 3$); (iv) imiloxan (3100 μg ; $n = 3$); and (v) JP-1302 (310 μg ; $n = 3$).

The last group ($n = 28$) received the above intracarotid infusions of capsaicin (control curve). Then, this group was subdivided into 7 subgroups ($n = 4$ each) that received intrathecal bolus injections (0.5 ml) of, respectively: (i) vehicle (saline); (ii) rauwolsine (310 μg); (iii) BRL44408 (310 μg); (iv) BRL44408 (1000 μg); (v) imiloxan (3100 μg); (vi) JP-1302 (100 μg); and (vii) JP-1302 (310 μg). After 10 min, each subgroup received an intrathecal bolus injection of B-HT 933 (3100 μg). Ten minutes later, the responses to the above intracarotid infusions of capsaicin were elicited again.

2.3. Other procedures applying to all protocols

Each dose of capsaicin, α -CGRP and acetylcholine was administered intracarotidly in a solution at a rate of 1 ml/min during 1 min. The dose intervals between the different doses of these compounds ranged between 5 min (acetylcholine) and 25 min (capsaicin and α -CGRP), as in each case we waited until the external carotid blood flow had returned to baseline values. The doses of these compounds were selected on the basis of results previously published (Jiménez-Mena et al., 2006), in which dose-dependent increases in external carotid blood flow were elicited with no changes in blood pressure or heart rate.

The correctness of the intrathecal administration of the different compounds was validated at the end of all experiments (after sodium pentobarbitone overdose) by injecting an intrathecal bolus injection of 1% methylene blue (0.5 ml), as reported by Muñoz-Islas et al. (2009). Then, the spinal cord was dissected by laminectomy at the level of the C₁–C₄ intervertebral discs to look for the catheter tip and dye distribution at this level.

Based on previous findings in mongrel dogs reported by our group using identical experimental methods (Muñoz-Islas et al., 2009), the Ethical Committee of our institution (CICUAL) approved all the experimental protocols of the present study as well as the number of dogs used for each group/subgroup (see Section 2.2 above). CICUAL also followed the regulations established by the Mexican Official Norm

(NOM-062-1999), in accordance with the NIH Guide for the Care and Use of Laboratory Animals.

2.4. Drugs

Apart from the anaesthetic (sodium pentobarbitone), the compounds used in this study were: L-phenylephrine hydrochloride, capsaicin, rat α -CGRP, acetylcholine chloride, 2-amino-6-ethyl-4,5,7,8-tetrahydro-6H-oxazolo-[5,4-d]-azepin-dihydrochloride (B-HT 933), imiloxan hydrochloride and rauwolsine hydrochloride (Sigma Chemical Co., St. Louis, MO, U.S.A.); 2-[(4,5-dihydro-1H-imidazol-2-yl)methyl]-2,3-dihydro-1-methyl-1H-isoindole maleate (BRL44408) (Tocris Cookson Inc., Ellisville, MO, U.S.A.); and acridin-9-yl-[4-(4-methylpiperazin-1-yl)-phenyl]amine (JP-1302) (gift from Dr. Jukka Sallinen; Orion Corporation ORION PHARMA, Turku, Finland). All compounds were dissolved in saline. This vehicle had no effect (when given intrathecally or intracarotidly) on the systemic and carotid haemodynamic variables. The doses of the antagonists refer to their respective salts, whilst those of the agonists refer to their free base.

2.5. Data presentation and statistical evaluation

All data are presented as mean $\pm$ S.E.M. The external carotid conductance was calculated by dividing the blood flow (ml/min) by the mean blood pressure (mm Hg) and multiplied by hundred. The peak changes in external carotid conductance were expressed as percent change from baseline. The difference between the variables within one group (or *different groups) of animals was compared by using a two-way repeated measures analysis of variance (or, correspondingly, *a two-way analysis of variance) (both randomized block design) followed by the Student–Newman–Keuls' test. Statistical significance was accepted at $P < 0.05$ (two-tailed).

3. Results

3.1. Systemic and carotid haemodynamic effects of the different treatments

Baseline values of mean blood pressure, heart rate and external carotid conductance in the 66 dogs were: 125 ± 18 mm Hg, 179 ± 26 bpm and 151 ± 28 ml/min mm Hg. Twenty minutes after starting the phenylephrine infusion a significant (and sustained) decrease in external carotid conductance (roughly 50%) was induced without significant changes in mean blood pressure or heart rate (Table 2). This procedure was systematically performed to enhance the vasodilator responses to capsaicin (Jiménez-Mena et al., 2006). Under these

Table 2

Values of mean arterial blood pressure (MAP; mm Hg), heart rate (HR; bpm) and external carotid conductance (ECC; ml/min mm Hg) in the 66 mongrel dogs before and 20 min after the continuous infusion of phenylephrine (1.5–6.8 $\mu\text{g}/\text{min}$), and in three different groups before and after: (i) 20 min the intracarotid infusion of phenylephrine had commenced (1.5–6.8 $\mu\text{g}/\text{min}$); and (ii) 10 min after intrathecal bolus injections of physiological saline (0.5 ml; given twice) or B-HT 933 (310 and 1000 μg given cumulatively; or 1000 and 3100 μg given cumulatively) during phenylephrine infusions.

		MAP (mm Hg)		HR (bpm)		ECC (ml/min mm Hg)	
Treatment	n	Before	After	Before	After	Before	After
Phenylephrine	66	125 ± 18	124 ± 20	179 ± 26	169 ± 23	151 ± 28	104 ± 27^a
Phenylephrine	4	113 ± 20	124 ± 20	175 ± 22	169 ± 22	187 ± 56	104 ± 27^a
Saline 1st (0.5 ml)		122 ± 20	131 ± 22	162 ± 15	160 ± 16	88 ± 13	84 ± 14
Saline 2nd (0.5 ml)		120 ± 23	119 ± 22	174 ± 21	171 ± 22	86 ± 15	99 ± 18
Phenylephrine	6	125 ± 8	143 ± 14	172 ± 8	159 ± 16	147 ± 9	82 ± 12^a
B-HT 933 (310 μg)		134 ± 11	135 ± 9	162 ± 8	165 ± 8	104 ± 10	82 ± 6
B-HT 933 (1000 μg)		135 ± 11	121 ± 12	157 ± 8	146 ± 7	111 ± 16	87 ± 11
Phenylephrine	4	92 ± 14	104 ± 12	179 ± 10	164 ± 8	194 ± 31	104 ± 19^a
B-HT 933 (1000 μg)		103 ± 13	98 ± 8	171 ± 4	160 ± 4	91 ± 25	95 ± 17
B-HT 933 (3100 μg)		102 ± 11	114 ± 34	157 ± 13	161 ± 11	93 ± 16	85 ± 17

^a $P < 0.05$, after vs. before from the corresponding baseline value. All values are expressed as mean $\pm$ S.E.M.

Table 3

Values of mean arterial blood pressure (MAP; mm Hg), heart rate (HR; bpm) and external carotid conductance (ECC; ml/min mm Hg) before and 10 min after the intrathecal bolus injections of physiological saline (0.5 ml), B-HT 933 (1000 and 3100 µg), rauwolsine (310 µg), BRL44408 (310 and 1000 µg), imiloxan (3100 µg) or JP-1302 (100 and 310 µg). Phenylephrine was infused exactly as described in Table 2.

Treatment	n	MAP (mm Hg)		HR (bpm)		ECC (ml/min mm Hg)	
		Before	After	Before	After	Before	After
Saline (0.5 ml)	16	130 ± 10	131 ± 10	163 ± 9	169 ± 11	105 ± 25	107 ± 25
B-HT 933 (1000 µg)	6	135 ± 11	121 ± 12	157 ± 8	146 ± 7	111 ± 16	87 ± 11
B-HT 933 (3100 µg)	8	110 ± 14	128 ± 17	164 ± 7	161 ± 6	89 ± 13	76 ± 15
Rauwolsine (310 µg)	7	107 ± 10	120 ± 11	165 ± 21	166 ± 23	128 ± 17	97 ± 12
BRL44408 (310 µg)	4	141 ± 10	134 ± 10	176 ± 25	160 ± 22	99 ± 19	96 ± 17
BRL44408 (1000 µg)	7	126 ± 9	127 ± 15	182 ± 19	184 ± 15	87 ± 7	85 ± 11
Imiloxan (3100 µg)	7	139 ± 11	141 ± 10	198 ± 21	197 ± 21	73 ± 5	68 ± 3
JP-1302 (100 µg)	4	164 ± 3	153 ± 4	156 ± 7	146 ± 4	74 ± 11	77 ± 8
JP-1302 (310 µg)	7	169 ± 2	164 ± 3	159 ± 6	156 ± 7	76 ± 8	85 ± 12

conditions, the above variables were not significantly modified in the animals that received intrathecal injections of: (i) vehicle (saline; 0.5 ml; given twice); (ii) B-HT 933 (310–3100 µg) (see Table 2); or (iii) the antagonists rauwolsine (310 µg), BRL44408 (310 and 1000 µg), imiloxan (3100 µg) and JP-1302 (100 and 310 µg) (see Table 3).

3.2. Effect of B-HT 933 on the external carotid vasodilator responses to capsaicin

The intracarotid infusions of capsaicin dose-dependently increased the external carotid conductance (Fig. 1A). These responses were: (i) unchanged after intrathecal vehicle (Fig. 1A); and (ii) dose-dependently inhibited after intrathecal B-HT 933 (1000–3100 µg) (Fig. 1B, C), with no significant changes in mean blood pressure or heart rate (Table 2). Since 3100 µg B-HT 933 inhibited the vasodilator responses to 31 and 56 µg/min capsaicin, this dose of B-HT933 was chosen for further pharmacological analysis.

3.3. Effect of an intrathecal bolus injection of saline or B-HT 933 on the external carotid vasodilator responses to capsaicin, α-CGRP and acetylcholine

The infusions of capsaicin, α-CGRP and acetylcholine dose-dependently increased the external carotid conductance (Fig. 2;

control responses) without significant changes in mean blood pressure or heart rate (not shown); these responses remained unchanged after intrathecal administration of physiological saline (Fig. 2A, B and C), as previously reported (Jiménez-Mena et al., 2006). The vasodilator responses to capsaicin and α-CGRP were longer-lasting (between 5 and 25 min) than those to acetylcholine (between 1 and 5 min). Interestingly, 3100 µg B-HT 933 (given intrathecally) significantly inhibited the vasodilator responses to capsaicin, particularly at 31 and 56 µg/min (Fig. 2D), but not those to α-CGRP (Fig. 2E) and acetylcholine (Fig. 2F).

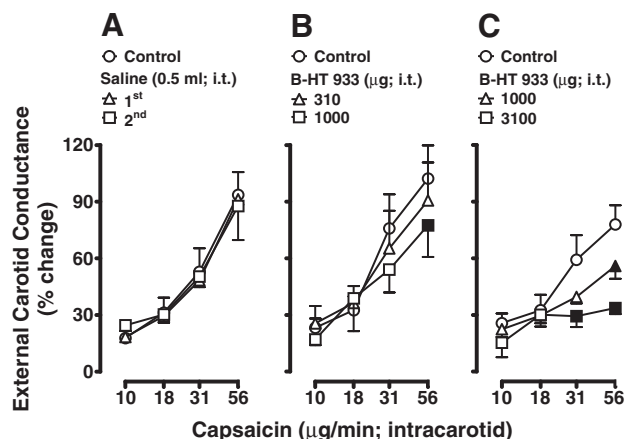


Fig. 1. Canine external carotid vasodilator responses to intracarotid infusions of capsaicin before (control) and after intrathecal bolus injections of either: (A) saline (0.5 ml given twice; $n=4$); (B) B-HT 933 (310 and 1000 µg; $n=6$); or (C) B-HT 933 (1000 and 3100 µg; $n=4$). For the sake of clarity, empty symbols depict either control responses (○) or non-significant responses (△) vs. control. Solid symbols (▲) represent significantly different responses ($P<0.05$) vs. control. Note that the control responses in (A) did not significantly differ from those in (B) and (C) ($P>0.05$).

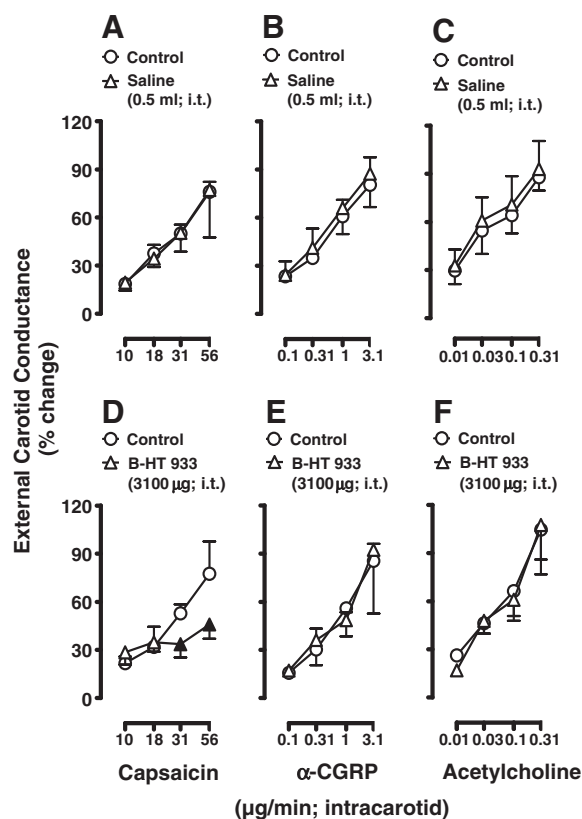


Fig. 2. Canine external carotid vasodilator responses to intracarotid infusions of capsaicin, α-CGRP and acetylcholine before (control) and after intrathecal bolus injections of either: (i) 0.5 ml vehicle (saline, $n=4$; A, B and C); or 3100 µg B-HT 933 ($n=4$; D, E and F). For the sake of clarity, empty symbols depict either control responses (○) or non-significant responses (△) vs. control. Solid symbols (▲) represent significantly different responses ($P<0.05$) vs. control.

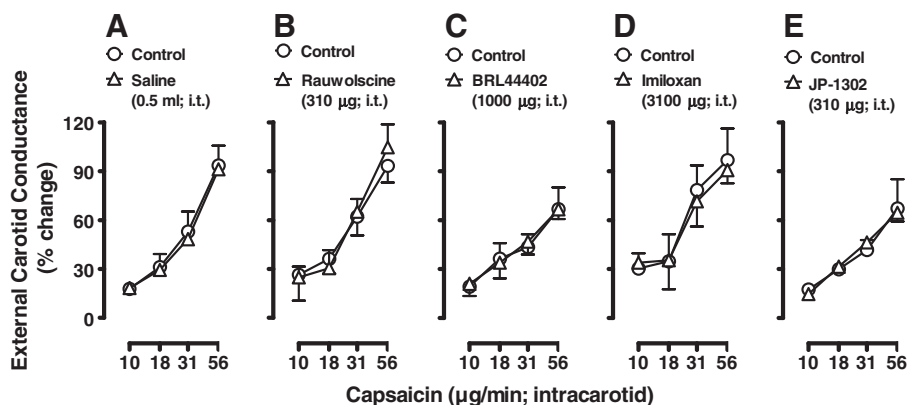


Fig. 3. Canine external carotid vasodilator responses to intracarotid infusions of capsaicin before (control) and after intrathecal bolus injections of: (A) vehicle (saline, 0.5 ml; $n=4$); (B) rauwolscline (310 µg; $n=3$); (C) BRL44408 (1000 µg; $n=3$); (D) imiloxan (3100 µg; $n=3$); or (E) JP-1302 (310 µg; $n=3$).

3.4. Effect of intrathecal bolus injections of vehicle, rauwolscline, BRL44408, imiloxan or JP-1302 on the external carotid vasodilator responses to capsaicin and on B-HT 933-induced inhibition of the external carotid vasodilatation to capsaicin

As shown in Fig. 3, intrathecal administration of vehicle (0.5 ml saline; Fig. 3A), rauwolscline (310 µg; Fig. 3B), BRL44408 (1000 µg; Fig. 3C), imiloxan (3100 µg; Fig. 3D) or JP-1302 (310 µg; Fig. 3E) failed to significantly modify *per se* the vasodilator responses to capsaicin. Hence, the intrathecal doses of these compounds are appropriate for identifying the pharmacological profile of B-HT 933-induced inhibition. Accordingly, the inhibition by B-HT 933 on capsaicin-induced external carotid vasodilatation was: (i) abolished by 310 µg rauwolscline, 1000 µg BRL44408 or 310 µg JP-1302 (Fig. 4B, D and G, respectively); (ii) partially blocked 310 µg BRL44408 or 100 µg JP-1302 (Fig. 4C and F, respectively); and (iii) unaffected by 3100 µg imiloxan (Fig. 4E) or an equivalent volume of vehicle (0.5 ml physiological saline; Fig. 4A) ($P>0.05$).

4. Discussion

4.1. General

Apart from the implications discussed below, our results show that direct spinal (C_1 – C_3) administration of B-HT 933 inhibits the external carotid vasodilatation to capsaicin. This response: (i) involves spinal α_2 -adrenoceptors (blocked by rauwolscline) that closely resemble the pharmacological profile of α_{2C} -adrenoceptors (potently blocked by JP-1302) and, to a lesser extent, of α_{2A} -adrenoceptors (moderately blocked by BRL44408); and (ii) seems unrelated to α_{2B} -adrenoceptors (resistant to imiloxan).

4.2. Systemic and carotid haemodynamic changes by the different treatments

As previously stated, intracarotid phenylephrine was continuously infused to enhance the vasodilator responses to capsaicin, α -CGRP and acetylcholine. This produced a sustained external carotid vasoconstriction (with no significant changes in blood pressure or heart rate; Table 2) due to the activation of α_1 -adrenoceptors, as previously shown in the canine external carotid bed (Willems et al., 2001a). Under these conditions, intrathecal vehicle (saline), B-HT 933 or the antagonists did not affect *per se* the baseline values of mean blood pressure, heart rate or external carotid conductance (see Tables 2 and 3).

Besides, the external carotid vasodilator responses to capsaicin, α -CGRP and acetylcholine were highly reproducible as they remained

essentially unchanged after intrathecal saline (Fig. 2), as previously reported by Muñoz-Islas et al. (2009).

As discussed earlier in similar studies, the mechanisms involved in the above vasodilator responses may involve: (i) capsaicin-induced CGRP release from perivascular nerves and, consequently, activation of CGRP receptors, as shown in the porcine carotid circulation (Kapoor et al., 2003a); (ii) activation of vascular CGRP receptors by exogenous α -CGRP, as shown in pigs (Kapoor et al., 2003b); and (iii) atropine-sensitive muscarinic receptors located on vascular endothelium for acetylcholine, as shown in the canine carotid circulation (Villalón et al., 1993). Although it is reasonable to speculate

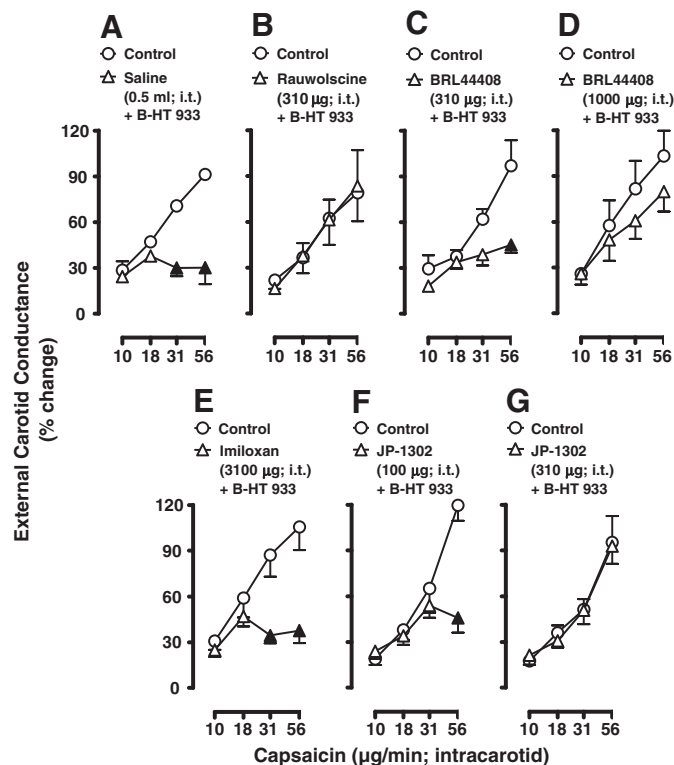


Fig. 4. Canine external carotid vasodilator responses to intracarotid infusions of capsaicin before (control) and after intrathecal B-HT 933 in dogs pretreated with an intrathecal bolus injection of: (A) vehicle (saline, 0.5 ml; $n=4$); (B) rauwolscline (310 µg; $n=4$); (C) BRL44408 (310 µg; $n=4$); (D) BRL44408 (1000 µg; $n=4$); (E) imiloxan (3100 µg; $n=4$); (F) JP-1302 (100 µg; $n=4$); or (G) JP-1302 (310 µg; $n=4$). For the sake of clarity, empty symbols depict either control responses (○) or non-significant responses (△) vs. control. Solid symbols (▲) represent significantly different responses ($P<0.05$) vs. control.

from our results that capsaicin-induced external carotid vasodilatation may also be mediated by CGRP release from perivascular sensory nerves, this hypothesis remains unproven. Clearly, further experiments will be required to corroborate this assumption.

4.3. Specific inhibition by spinal B-HT 933 of the vasodilator responses to capsaicin

The fact that only the vasodilatation to capsaicin was inhibited by intrathecal B-HT 933 (Fig. 2) suggests that this response: (i) is highly specific; and (ii) involves a central (spinal) mechanism probably related to inhibition of trigeminal neurotransmission rather than interactions with vascular CGRP receptors and/or muscarinic receptors. Although it could still be argued that the inhibition by B-HT 933 was due to its leakage into the circulation, this seems unlikely since intrathecal B-HT 933 did not affect mean blood pressure, heart rate or external carotid conductance (Tables 2 and 3). Thus, the site of action of B-HT 933 is mainly spinal rather than peripheral. Admittedly, the intrathecal doses B-HT 933 (310–3100 μ g) seem very high, as the cerebrospinal fluid in mongrel dogs is about 10–30 ml. However, one should keep in mind: (i) the moderate affinity of B-HT 933 for α_2 -adrenoceptors (pK_i : 7.0); (ii) the deep anaesthesia of the animals; and (iii) that similar studies have used high intrathecal doses of other compounds in dogs (Horais et al., 2003; Lida et al., 1997).

4.4. The role of spinal α_2 -adrenoceptors in the inhibition of the external carotid vasodilator responses to capsaicin

In order to identify the pharmacological profile of the receptors involved in the spinal inhibition by B-HT 933 of the carotid vasodilator responses to capsaicin, it was of the foremost importance to ascertain firstly that the intrathecal administration of the α_2 -adrenoceptor antagonists rauwolscine, BRL44402, imiloxan or JP-1302 was devoid of effects *per se* on: (i) the vasodilator responses to capsaicin (Fig. 3); or (ii) the baseline values of mean blood pressure, heart rate and external carotid conductance (Table 3). Accordingly, any effect of a given antagonist on the inhibition by B-HT 933 is due to a direct interaction of the antagonist with its spinal receptor.

As shown in Fig. 4, the fact that B-HT 933-induced inhibition was abolished by 310 μ g rauwolscine (which displays high affinity for all α_2 -adrenoceptor subtypes; Table 1), shows the role of α_2 -adrenoceptors. For elucidating the role of α_{2A} -, α_{2B} -, and α_{2C} -adrenoceptor subtypes in the inhibition by B-HT 933, the antagonists BRL44408 (α_{2A}), imiloxan (α_{2B}) and JP-1302 (α_{2C}) were used in doses that, based on their affinities (Table 1), are enough to block their respective subtypes. Since B-HT 933-induced inhibition was dose-dependently blocked by 310 and 1000 μ g BRL44408 or 100 and 310 μ g JP-1302, remaining unchanged after 3100 μ g imiloxan (Fig. 4), the simplest interpretation suggests that mainly $\alpha_{2A/2C}$ -adrenoceptors are involved. Nevertheless, the profile of blockade produced by the above doses of BRL44408 and JP-1302, in conjunction with their affinities and selectivity (Table 1), deserves further considerations.

Accordingly, if α_{2A} -adrenoceptors played a major role, 310 μ g BRL44408 (pK_i for α_{2A} : 8.2) should have abolished (instead of partly blocking; Fig. 4C) B-HT 933-induced inhibition. Interestingly, 1000 μ g BRL44408 practically abolished this response (Fig. 4D), but this effect cannot be attributed to exclusive blockade of α_{2A} -adrenoceptors since: (i) BRL44408 also displays some affinity for α_{2B} (pK_i : 6.2) and α_{2C} (pK_i : 6.8) adrenoceptors; and (ii) the *in vitro* selectivity of BRL44408 for α_{2A} -adrenoceptors vs. α_{2C} -adrenoceptors is small (Table 1), leaving little room for *in vivo* selectivity.

Moreover, the fact that B-HT 933-induced inhibition: (i) remained unchanged after 3100 μ g imiloxan (Fig. 4E), an α_{2B} -adrenoceptor antagonist (pK_i α_{2B} : 6.9; Table 1), excludes the role of α_{2B} -adrenoceptors; and (ii) was partly blocked after 100 μ g JP-1302 (Fig. 4F) and abolished

after 310 μ g JP-1302 (Fig. 4G) (a selective α_{2C} -adrenoceptor antagonist; Sallinen et al., 2007), suggests a major role for α_{2C} -adrenoceptors. Consistent with this suggestion, the affinity of JP-1302 for α_{2A} (pK_i : 5.5) and α_{2B} (pK_i : 5.8) adrenoceptors is about 2-log units lower than that for α_{2C} -adrenoceptors (pK_i : 7.5) (Table 1). Indeed, the affinities and selectivity of JP-1302 clearly contrast with those of the less selective α_{2C} -adrenoceptor antagonist MK912 (Table 1). Hence, activation of α_{2C} -adrenoceptors seems to play an important role under our experimental conditions, unlike the conclusion (involvement of $\alpha_{2A/2C}$ -adrenoceptors using i.v. clonidine and MK912) by Jiménez-Mena et al. (2006). Although no subtype selective α_{2C} -adrenoceptor agonist is available yet, the advent of such a compound would allow us to further corroborate the role of spinal α_{2C} -adrenoceptors in our experimental model.

Admittedly, the above interpretations are based on the assumption that species differences between the binding of B-HT 933, rauwolscine, BRL44402, imiloxan and JP-1302 to canine and human α_{2A} -, α_{2B} - and α_{2C} -adrenoceptors do not play a major role.

4.5. The modulatory role of spinal α_2 -adrenoceptors in dogs: resemblance to the function of other α_2 -adrenoceptors

Our findings showing the modulatory role spinal α_2 -adrenoceptors in dogs are consistent with other studies. For example, electrophysiological experiments in neurons from the rat trigeminal nucleus caudalis have shown that spinal activation of α_2 -adrenoceptors inhibited the N-methyl-D-aspartic acid (NMDA)-evoked responses (Zhang et al., 1998). Consequently, blockade of NMDA receptors with ketamine or MK801 inhibited the dural vasodilatation to capsaicin and dural electrical stimulation in an intravital microscopy model (Chan et al., 2010). This reinforces the role of spinal mechanisms modulating peripheral vascular responses. Moreover, Takeda et al. (2002) have shown that α_2 -adrenoceptor agonists hyperpolarize trigeminal root ganglion neurons. Thus, activation of α_2 -adrenoceptors may inhibit the nociceptive transmission in the trigeminal system at the level of both trigeminal root ganglion neuronal bodies and primary afferent terminals.

4.6. Possible clinical implications

Over the years, it has been suggested that vasodilatation of the extracranial terminal branches of the external carotid artery are a significant source of pain in migraine (Graham and Wolff, 1938; Saxena and De Vlaam-Schluter, 1974; Shevel, 2011a, 2011b). Indeed, antimigraine drugs such as the triptans (Muñoz-Islas et al., 2006; Villalón et al., 1995) and ergots (Villalón et al., 1999) produce vasoconstriction in the canine external carotid circulation. Additionally, the central actions of triptans result in the inhibition of the external carotid vasodilatation to capsaicin (Muñoz-Islas et al., 2006, 2009).

Hence our results lead us to suggest that an agonist with selectivity at central α_{2C} -adrenoceptors may have potential antimigraine properties. Such an agonist would be devoid of cardiovascular effects, as α_{2C} -adrenoceptors do not play a role in cardiovascular regulation (Link et al., 1996; MacMillan et al., 1996). Obviously, the clinical use of antimigraine drugs which do not cross the blood-brain barrier is more convenient as this approach can prevent unwanted central side effects. Moreover, our findings may shed further light on the mechanisms involved in the antimigraine efficacy of some agents with complex pharmacology including ergotamine (Valdivia et al., 2004), dihydroergotamine (Villalón et al., 2004) and isometheptene (Willems et al., 2001b), which produce canine external carotid vasoconstriction via $\alpha_{2A/2C}$ -adrenoceptors. Thus, their antimigraine actions within the bounds of α_2 -adrenoceptor activity could be mainly mediated by cranial vasoconstriction and, to a certain extent, by the inhibition of the trigeminovascular complex.

In view that CGRP plays a role in the pathophysiology of migraine (see Introduction section), CGRP receptor antagonists have been shown to be effective in the acute treatment of migraine without producing cranial vasoconstriction or the vascular side effects typical of the triptans and ergots (Villalón and Olesen, 2009).

5. Conclusion

Intrathecal B-HT 933 inhibited the carotid vasodilatation to capsaicin. This response, mediated by spinal α_2 -adrenoceptors unrelated to the α_{2B} -subtype, resembles the pharmacological profile of α_{2C} -adrenoceptors and, to a lesser extent, α_{2A} -adrenoceptors.

Acknowledgements

The authors would like to thank Mr. Mauricio Villasana for his skilful technical help. We are also indebted to CONAcYT (Mexico) for their financial support (grant 60789-M), and the pharmaceutical companies for their generous gifts (see Drugs section).

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